

Lab 3.1: Build a Reusable Component Set

|
--- | ---

****Goal**** | Create generic `Button` and `Card` components that use props, defaults and children.

****Suggested time**** | 40 min

You are building your own mini design system. These two components will be reused throughout TaskBoard.

Steps

1. Create `src/components/ui/Button.jsx`.
2. Give it the props `variant` ("primary" or "secondary", default "primary"), `size` and `children`.
3. Build the `className` from the props with a template literal.
4. Create `src/components/ui/Card.jsx` with `title` and `children` (if you built `Card` during Module 3.2, you already have it).
5. Add CSS for `.btn-primary`, `.btn-secondary` and `.card` in `src/App.css`. If you removed the stylesheet import on Day 2, add `import './App.css'`; back at the top of `App.jsx`.
6. Render a `Card` containing two `Buttons` in `App`.
7. Inspect both components in React DevTools.
8. Commit: `git add .` then `git commit -m "Add Button and Card UI components"`.

Done when

- `Button` renders both variants
- `Card` shows a title only when one is passed
- There are no console warnings

Hints (try on your own first)

`src/components/ui/Button.jsx`

```
function Button({ variant = "primary", size = "md", children }) {  
  return <button className={`btn btn-${variant} btn-${size}`}>{children}</button>;  
}  
export default Button;
```

`src/App.jsx`

```
import './App.css';  
import Button from './components/ui/Button.jsx';  
import Card from './components/ui/Card.jsx';  
  
function App() {  
  return (  
    <main>  
      <Card title="Sprint 12">  
        <Button>Save</Button>  
        <Button variant="secondary">Cancel</Button>  
      </Card>
```

```
<Card>
<p>This card has no title.</p>
</Card>
</main>
);
}
export default App;
```

Troubleshooting

Problem | Fix

--- | ---

Styles not applied | `App.css` is not imported in `App.jsx`, or the class names do not match the CSS.

`Objects are not valid as a React child` | You passed an object as children, for example `{task}`. Render a property such as `{task.title}` instead.

Component renders nothing | The function has braces but no `return`.

Whole page blank; the Console says the module `does not provide an export named 'default'` | The component file is missing `export default`.

> Stretch challenge

> Accept any extra props with rest, and spread them onto the button element so that onClick and disabled work automatically:

```
>
> jsx
> function Button({ variant = "primary", size = "md", children, ...rest }) {
>   return (
>     {children}
>   );
> }
>
```